

Technical Section

EPCS: Endpoint-based part-aware curve skeleton extraction for low-quality point clouds

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ABSTRACT

The curve skeleton is an important shape descriptor which has been utilized in various applications in computer graphics, machine vision, and artificial intelligence. In this study, the endpoint-based part-aware curve skeleton (EPCS) extraction method for low-quality point clouds is proposed. The novel random center shift (RCS) method is first proposed for detecting the endpoints on point clouds. The endpoints are used as the initial seed points for dividing each part into layers, and then the skeletal points are obtained by computing the center points of the oriented bounding box (OBB) of the layers. Subsequently, the skeletal points are connected, thus forming the branches. Furthermore, the multi-vector momentum-driven (MVMD) method is also proposed for locating the junction points which connect the branches. Due to the shape differences between different parts on point clouds, the global topology of the skeleton is finally optimized by removing the redundant junction points, re-connecting some branches using the proposed MVMD method, and applying an interpolation method based on the splitting operator. Consequently, a complete and smooth curve skeleton is achieved. The proposed EPCS method is compared with several state-of-the-art methods, and the experimental results verify its robustness and effectiveness. Furthermore, the skeleton extraction and model segmentation results on challenging point clouds of broken Terracotta also highlight the utility of the proposed method.

1. Introduction

The curve skeleton is a typical shape descriptor for 3D models that concisely and intuitively reveals the geometry, topology, and symmetry structure of the described shape [1]. Curve skeletons play a critical role in shape completion [2], shape segmentation [3,4] and model classification and recognition [5]. The extraction of the curve skeleton is also of great significance for the restoration of broken cultural relics, especially for symmetry models like vases. Moreover, the skeleton can be employed to reconstruct the surfaces of cultural relics with missing fragments [2,6,7]. Curve skeletons are most widely used because they contain concise topology information about objects [2,5,6,8–11]. This study focuses on the extraction of the curve skeleton by taking into account both the shape of the object and the topological structure between its parts.

Raw scan point clouds are usually obtained via reconstruction from a series of 2D images [12,13] or scanning with a 3D scanner [14]. Due to many limitations of the scanning process and the raw scan points are usually disorganized, lack inherent structural or directional information, and contain noise, and outliers [15]. Furthermore, because of the occlusions, raw scan point clouds generally have holes. As a result, raw scan point clouds are usually of low-quality, especially

those without any processing such as noise reduction or hole filling. Therefore, the algorithms need to be robust to highly non-uniform sampling patterns, diverse levels of noise, and even missing data [2].

A typical strategy for extracting skeletons is based on a Voronoi diagram (VD), but the computation of the VD is complex and computational expensive [1,16]. Alternatively, curvature flow-based methods have also achieved satisfactory performance [17–19]; however, these methods are time-consuming because they require the computation of the curvature and other geometric invariants, and they can only handle dense point clouds without missing data and noise. The third category of methods exploits some geometric features of the skeleton, such as the rotational symmetry axis (ROSA) [2], and the local separators [20], which can be applied to the raw scan point clouds. Several state-of-the-art methods utilize a robust central axis to fit each part of the model and iteratively obtain the curve skeleton, such as ROSA [2], L_1 -median [6]. These methods usually take a resampled model as the input to reduce the computational cost, such as [6,10]; however, they may produce incomplete skeletons. To tackle this problem, in the present study, all parts of the point cloud are first located by detecting the endpoints of the parts, and the skeletal curve of each part is regarded as the branch of the final skeleton. Furthermore, the essence of skeletonization is to

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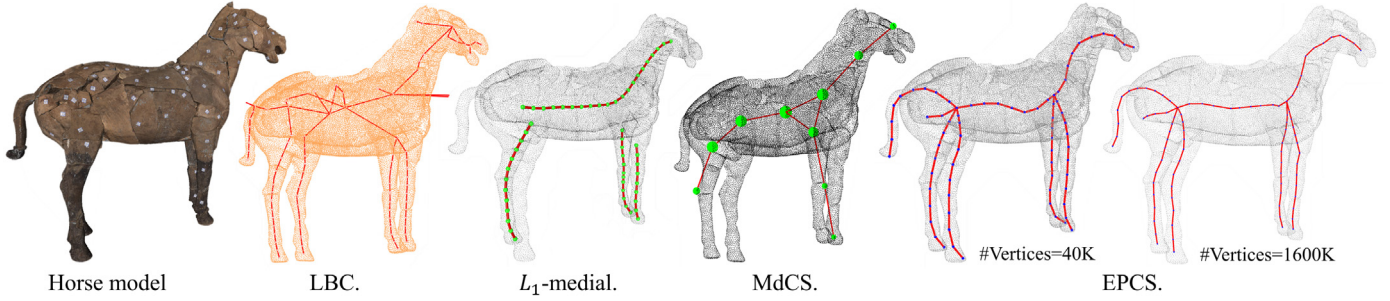


Fig. 1. The results of the LBC [9], L_1 -medial [6], MdCS [10] and EPCS methods on the horse model from the Terracotta dataset, which is a point cloud with missing data caused by holes and missing pieces of Terracotta, the model is constructed by reassembling.

appropriately divide the part into layers [10,21,22], and the center point of each layer is considered as a skeletal point.

Therefore, the endpoint-based part-aware curve skeleton (EPCS) extraction algorithm is proposed, and Fig. 1 presents the results of the EPCS algorithm on a raw scan point cloud. First, the novel random center shift (RCS) method is introduced to detect the endpoints, which are utilized as the seed points for the skeleton generation. Subsequently, region growing [23], is used to guide the division of layers, and the center points of the oriented bounding boxes (OBBs) [24] of the layers are then regarded as the skeletal points. Because layer division is a linear sequential process, the skeletal curves can be directly obtained by connecting the skeletal points in order. However, the skeletal curves obtained in this stage are independent branches; thus, the multi-vector momentum-driven (MVMD) method is proposed, which considers both the local distance and direction information and iteratively connects the branches, thus forming the complete skeleton. Because global information about the given point cloud is not considered when connecting the branches, and the distribution of skeletal points is non-uniform and not smooth due to the shape differences between different parts, the global topology of the skeleton is finally optimized by: removing the redundant junction points, re-connecting some branches using the proposed MVMD method, and applying an interpolation method based on the 1-4 splitting operator [25] to smooth the skeletal curves. Consequently, a complete and smooth curve skeleton is obtained.

The main contributions of this study are summarized as follows:

- A novel EPCS method is proposed for curve skeleton extraction, which is robust to noises, outliers, and missing data, and can be applied to low-quality point clouds. The proposed EPCS requires no preprocessing, for example, mesh reconstruction, Voronoi graph generation, or the computation of geometric invariants.
- A novel RCS method is proposed for the detection of the endpoints on point clouds, which can effectively avoid missing branches during skeleton extraction.
- The MVMD method is proposed for locating the junction points which connect the branches, and this also significantly improves the completeness and smoothness of the skeletal curves in the connected regions.
- The proposed EPCS outperforms current state-of-the-art methods in terms of effectiveness and robustness.

2. Related work

According to Tagliasacchi et al. [1], there are three types of skeletons, namely: surface skeletons [26–28], curve skeletons [2,6], and centerlines [29–31]. Among the types of skeletons, curve skeletons are more likely to directly promote the restoration of cultural relics. Curve skeletons are 1D, and many ideas for extraction have been developed. In this section, we only review the methods of extracting curve skeleton. The input model for the curve skeleton extraction algorithm has two types: mesh and point cloud. Among them, the algorithms based on Voronoi diagram (VD) and curvature flow use mesh shape as input

to extract the curve skeleton by using the contraction principle [32], while other algorithms use the point cloud model as input.

In recent years, several methods to compute the curve skeleton from the mesh shape are using the contraction principle. The use of a Voronoi diagram (VD) to extract the skeleton is a typical idea. Ogniewicz et al. [33] proposed a skeletonization method based on the VD of boundary points. Subsequently, based on a given graph and a 3D polygonal surface, Lu et al. [16] used centroidal Voronoi tessellation (CVT) to achieve skeleton fitting and conduct model segmentation in a point cloud data set.

Alternatively, Au et al. [17] used curvature flow to achieve Laplacian smoothing and complete the geometry contraction process. Subsequently, Chuang et al. [18] published an approach for the tracking of a hierarchical finite element system with a surface evolving under mean curvature flow (MCF), which is characterized by a time advantage. Considering the skeletonization problem via MCF, Tagliasacchi et al. [34] generated mean curvature skeletons. Vahrenkamp et al. [19] also constructed the mean curvature skeleton to identify object regions that are suitable for grasping by robotic hands; the mean curvature skeleton was found to achieve good results and performance on watertight surface meshes. However, these methods can only be used on models that do not contain missing data and that are processed after scanning, and require the input point cloud to be clean and dense enough to generate the correct curve skeleton.

Subsequently, the following methods employ point cloud models as input. In recent years, the object of numerous skeleton extraction studies has been the tree model [35–39]. Livny et al. [35] presented an automatic approach that robustly reconstructs the skeletal structures of trees via the active laser scanning of real-world vegetation data. Fu et al. [36] extracted a skeleton from tree point clouds via an octree and the level set method. In contrast to the first two methods which employ unoriented point clouds as input, the following three methods utilize oriented point clouds. Wu et al. [37] extracted a skeleton to estimate the phenotypic parameters of maize plants with high accuracy. Bucksch et al. [38] report a new algorithm developed for the skeletonization of an incomplete point cloud obtained by laser scanner, it can be applied for obtaining botanic tree parameters like length and diameter of branches. Mei et al. [39] developed a novel method to effectively model trees from a single laser scan, and the experiment results demonstrate the effectiveness and robustness of it for coping with complex shapes of branching structures and occlusions.

Jalba et al. [40] proposed a method based on the GPU framework to extract the surface and curve skeletons of oriented point cloud models. Hu et al. [41] presented an approach for skeleton expression based on the set of the cross-section centroids from an oriented point cloud model. Zhou et al. [42] utilized the k-nearest neighbors (KNN) algorithm with an iterative point contraction algorithm to extract the curve skeleton. Several methods for improving the geometric properties and thus suitable for skeletonization have been proposed. Li et al. [43] improving the use of LOP (Locally Optimal Projection) to adaptively contract medial surfaces of 3D models for skeletonization.

Song et al. [44] used the distance field to guide the iterative process of L_1 -medial to obtain a better skeleton compared to [6].

Some algorithms that have a strong correlation with the proposed algorithm are subsequently discussed. These algorithms can extract the curve skeleton from the raw scan or missing data, and all have the ability to resist noise. Tagliasacchi et al. [2] proposed an algorithm based on the rotational symmetry axis (ROSA) of an oriented point set in 2009; this algorithm can extract the curve skeleton from incomplete point clouds in which large portions of the data may be missing. Inspired by the ROSA algorithm, Cao et al. [9] subsequently presented an algorithm for curve skeleton extraction via Laplacian-based contraction (LBC) in 2010. Because LBC is considered to be an improvement of ROSA, the proposed algorithm is only compared with LBC in this study. In 2013, Huang et al. [6] introduced the L_1 -medial skeleton (L_1 -medial) as a curve skeleton, and presented an iterative algorithm to obtain it. In 2020, Qin et al. [10] proposed a Mass-driven topology-aware curve skeleton (MdCS) that displays the mass distribution while retaining geometric properties. Subsequently, Jiang et al. [11] introduce a novel contraction method called “graph contraction” (GC) to contract 3D tree-shaped point clouds, which is computed by the minimization of an energy function. All five methods require point cloud sampling to improve computational efficiency, and ROSA [2] and LBC [9] additionally employ oriented point clouds as input.

Different from existing methods that consider the topology of the skeleton after the rough skeleton is obtained, such as [2,6,9,10], the proposed method preliminarily considers the skeletal branches via endpoint recognition, which effectively avoids missing branches. Since region growing can realize the layering of the model directly, and the layering results can be used as the input of skeletal point computation, region growing strategy has been widely used in the skeleton extraction task such as [43] and [45]. Meanwhile, a small radius constraint is used to enhance the region growing strategy, thus avoiding the influence of noises and outliers, e.g., the colored points in Fig. 2(n) are all the points which are visited in the region growing process; whereas the noises and outliers that are marked gray and far from the model are not involved in the calculation. Furthermore, the advantage of using the center point of OBB as the skeletal point is that OBB is insensitive to missing data, this is because even if the OBB is constructed by using the point set with missing data, the center point of the OBB will not change significantly; moreover, OBB is robust to noise and outliers [24] due to the computation of the principal directions of the point set when constructing the OBB, thus reducing the impact of noises and outliers on the bounding box construction. Therefore, the proposed EPCS performs well on the raw scan point cloud with holes, noises, and outliers.

In addition, although the L_1 -median is robust to noises and outliers, when applied to a raw scan point cloud with holes, it will be biased towards the opposite side of the holes. Therefore, the L_1 -medial skeleton [6] needs the re-centering process to decrease this deviation. In addition, the iterative computation of L_1 -medial also involves a high computational cost. Moreover, OBB and axis aligned bounding box (AABB) are the most commonly used bounding boxes [24] that can obtain the center point of the given model with missing data. However, the deviation of the center point of AABB may be induced if the model has not been aligned with the axes, whereas the OBB does not lead to such case. In this study, the center point of the OBB is considered as the skeletal point.

Summary. The proposed algorithm differs from these existing algorithms in the following aspects. During the computation process, the proposed algorithm requires the consideration of only the distance between points; it does not need to estimate the normal [2,9,10] or curvature [18,34], and does not require the construction of the connectivity between the points, like voxelization [46,47], and meshing [34,48]. Furthermore, while the ROSA [2] and LBC [9] methods depend on the assumption that the cross-sections of the input shape are

near-circular structures, the proposed algorithm has no such limitation. Furthermore, unlike the L_1 -medial [6] and MdCS [10] methods, the proposed EPCS algorithm utilizes the RCS method to detect the structure of the final curve skeleton via endpoint detection before obtaining the rough skeleton, thereby effectively avoiding the missing of skeletal branches.

3. Methodology

3.1. Overview

Given an unoriented point cloud $P = \{p_i\}_{i \in I} \subset \mathbb{R}^3$ the proposed EPCS method directly extracts the curve skeleton of P . EPCS method contains three stages. The pipeline of the proposed method is illustrated in Fig. 2. First, We use the RCS method to detect the endpoint. The endpoints of different parts of P are recognized via the RCS method, and are marked as small red balls in Fig. 2(c). Each endpoint corresponds to a part of the model, and the skeleton of this part is a branch of the skeleton. The endpoints are then used as the seed points for layer division.

In the second phase, skeletal points are generated. Each part of P is divided into several layers via region growing from the seed points, as shown in Fig. 2 (d–g). The skeletal points are then achieved by computing the centers of the OBB of each layer, as shown in Fig. 2(g). The branches of the skeleton can be obtained by connecting the skeletal points in sequence, as shown in Fig. 2(h). However, to construct a complete curve skeleton, the branches should be connected by the junction points.

The third stage constructs the curve skeleton. The MVMD method is proposed for both generating junction points and connecting the branches. Particularly, by considering both the local distance and direction information, junction points are generated as shown in Fig. 2(i). However, since the distance and direction information are local information whereas the global information are not considered, redundant junction points may also be generated; thus, the MVMD method is employed again to remove the redundant junction points, and thereby constructing a complete skeleton as shown in Fig. 2(j) and (k). Moreover, an interpolation method based on the splitting operator [25] is also used to smooth the obtained final skeletal curves, as shown in Fig. 2(l), and resample the skeletal points to ensure the uniform sampling of the skeletal points. Consequently, a complete and smooth curve skeleton of P can be obtained, as shown in Fig. 2(m). The result of model segmentation of the P is shown in Fig. 2(n).

3.2. RCS for endpoint detection

The skeletal curve of each part of the given point cloud is considered a branch of the final skeleton. To avoid missing branches, a skeletal curve of each part is first generated by detecting the endpoint of each part. In this study, the endpoints refer to the outermost sharp points of the parts of the point cloud, e.g., the endpoints of an ostrich model are the outermost sharp points on the toes, and head, as shown in Fig. 2(c). The observation of endpoints indicates that the center point of their neighbors quickly shifts with the large increase of the neighborhood radius, whereas the non-endpoints have no such characteristic, as shown in Fig. 3. Thus, the novel RCS method is proposed, which uses the shift distance of the center points of different scales of neighborhoods for endpoint recognition. This section details the proposed RCS method.

Given a scattered and unoriented point cloud P , all the points of P are normalized in range of $[0, 1]$ to avoid the influence of the scale of the given point cloud. Then, the RCS method is performed on the normalized model P , and the detailed process is as follows.

The RCS method starts with searching the points on the end regions of the parts of P . Specifically, for each point $p_i \in P$, its neighboring points are searched using two different scales of radii, i.e., when the search radius is r_1 , the set of neighbors is denoted as S_1 (the blue points

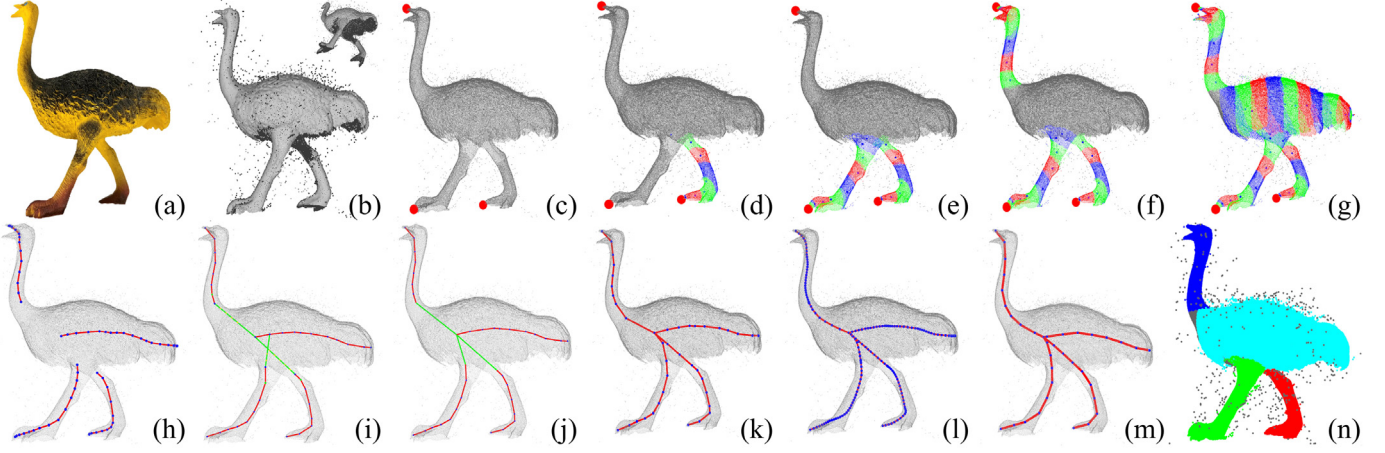


Fig. 2. The pipeline of the proposed method. Given an ostrich model (a) and its raw scan point cloud (b), the RCS method recognizes 3 endpoints marked as small red balls in (c). To extract the skeletal points of the right leg, the leg is divided into layers (d), which are marked in red, green, and blue; each layer corresponds to a skeletal point. Then, the same process is used to extract the skeletal points of the left leg (e), head (f) and body (g). The skeletal branches obtained are shown in (h). Junction points are computed to connect the branches (i-k). Then, to smooth the skeletal curves (l), an interpolation method is used to resample the skeletal points. Finally, this results in a complete and smooth curve skeleton (m). The model segmentation result is shown in (n).

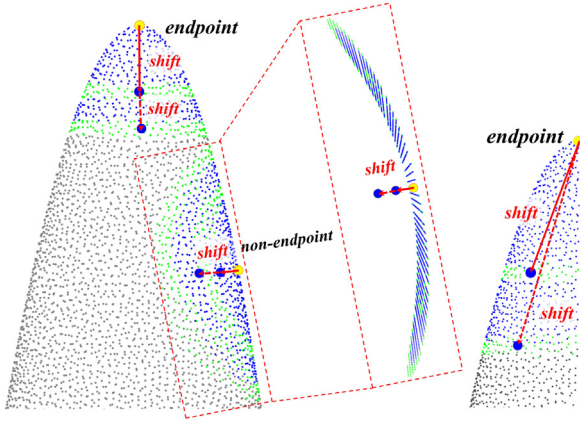


Fig. 3. The illustration of random center shift (RCS) method. The center point (blue ball) of the endpoint (yellow ball) neighbors quickly shifts with the large increase of the neighborhood radius. Left: complete model, right: incomplete model.

in Fig. 3), and when the search radius is r_2 , the set of neighbors is denoted as S_2 , where $r_1 = \delta\ell$, $r_2 = r_1 + \epsilon$, $\epsilon > 0$, ℓ is the diagonal length of the OBB of P . p_o is the center point of point set S_c and $S_c = \{p|p \in S_2, p \notin S_1\}$ (p_o is marked by a blue ball, S_c is the green points in Fig. 3). When the input model is incomplete, the RCS can still obtain effective endpoints. This is because the shift is also notably large at the endpoints, as shown in Fig. 3. Then, the point p_i that meets Eq. (1) constitutes the point set S_i , S_i contains the points located on the end regions, such as the yellow points in Fig. 4.

$$S_T = \{p_i | \|p_i - p_o\| + \epsilon > r_1\} \quad (1)$$

Avoiding erroneously regarding outliers as endpoints, it still requires estimating the local sampling density [49] of p_i .

It should be noted that the point set S_i contains all points located on the end regions of P ; thus, hierarchical clustering [50] is performed to divide S_i into different subsets, and each subset only contains points on the same part. For example, as shown in Fig. 4, the resulting clusters are the end region points on the four feet, the tail, and the head of the wolf, respectively. The clustering process is summarized in Algorithm 1. The parameter ϕ can prevent outliers from being considered as a subset of S_i .

The subsets of S_i are denoted as S^{part_i} , and $part_i$ represents the i th part of P . The endpoints are then computed. For each $p_p \in S^{part_i}$, its

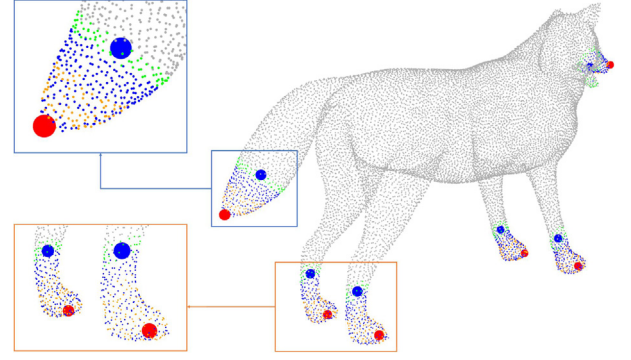


Fig. 4. The endpoints are recognized by the RCS method and marked with red balls. The blue balls are the center points of OBB of $S_T^{part_i}$.

neighboring points within radius ϵ are searched, and the neighboring point set is denoted $S_E^{part_i}$, e.g., the blue points in Fig. 4. Then, for each $p_e \in S_E^{part_i}$, its neighboring points within radius ϵ are also searched, and the neighboring point set is $S_T^{part_i}$, e.g., the green points in Fig. 4. To increase the robustness, the OBB of $S_T^{part_i}$ is then calculated, and the center point (the blue balls in Fig. 4) of the OBB is denoted as $p_c^{part_i}$ instead of directly computing the center points of $S_T^{part_i}$. Given $p_c^{part_i}$, the endpoint p_i^p of the i th part is defined as follows:

$$p_i^p = \max \|p_c^{part_i} - p_p\|, \quad (2)$$

where p_i^p is the point that has the largest Euclidean distance from $p_c^{part_i}$. The endpoints of the wolf model are marked with red balls in Fig. 4.

3.3. Skeletal points generation

Given the endpoints, the skeletal points of each part are generated in this stage. The endpoints of P were generated in the previous stage; however, it can be seen from Fig. 4 that the body part has no endpoints because it is surrounded by other parts, such as the tail and the legs. If the skeleton for each part is directly generated based on the endpoints, the skeleton of the body part will be missed; in addition, the generation of the skeleton of the body part from any endpoint is also unstable. Therefore, the skeleton of one part is generated in only one iteration, and the processed part will not be considered in the next iteration,

Algorithm 1 Euclidean distance cluster**Input:** point set: S_l , parameters: δ, ℓ, ϕ **Output:** S^{part}

```

1:  $S^{part} \leftarrow \emptyset$ 
2: for  $p \in S_l$  do
3:    $Q \leftarrow \emptyset$ 
4:   if  $p.status \neq processed$  then
5:      $Q.add(p)$ 
6:     for  $q \in Q$  do
7:        $q.status \leftarrow processed$ 
8:        $N = Knn_q(\delta\ell)$ 
9:        $Q.addAll(N)$ 
10:    if  $size(Q) \geq \phi$  then
11:       $S^{part}.add(Q)$ 
12:    end if
13:  end for
14: end if
15: end for

```

which deals with another part; this decreases the influence between each part.

Because the skeleton of one part is generated in only one iteration, the endpoints should first be sorted. The order of the endpoints determines the processing order of the parts and the generation order of the skeletal curves (referred to as branches). The endpoints are sorted in an increasing sequence of the adaptive radius r_i^p , which is the average distance from $p_c^{part_i}$ to each point in $S_T^{part_i}$.

After sorting the endpoints, each part is divided into layers via the classic region growing strategy, as shown in Fig. 2 (d–g). The entire layer division process consists of a two-layer loop; the outer loop selects an unvisited endpoint, and the inner loop computes the skeletal points of each part.

In the outer loop, the endpoint p_i^p that has not been visited is taken as the initial seed. Given the initial seed p_i^p , its neighboring points within radius r_i^p form the point set $S_{layer_1}^i$, which is the first layer of the i th part. Then, the center point c_1^i of the OBB of $S_{layer_1}^i$ is computed, and c_1^i is regarded as the first skeletal point of this branch.

Some pre-processing is then required by the inner loop. The average distance d_1^i between c_1^i and all points in $S_{layer_1}^i$ is computed, and all points in $S_{layer_1}^i$ are pushed into Q_{seed} , which stores the growing seed points. In the inner loop, each region begins from a growing seed p_{seed} that popped from Q_{seed} . Then, to obtain $S_{layer_{l+1}}^i$, the neighboring points of p_{seed} within radius Δ that are not labeled as “visited” form the point set $S_{layer_{l+1}}^i$, and the center point of the OBB of $S_{layer_{l+1}}^i$ is computed as a new skeletal point c_{l+1}^i . Finally, the points in $S_{layer_l}^i$ are labeled as “visited”, and no longer participate in the subsequent processing.

Furthermore, d_{l+1}^i is also computed, which controls the iteration of the inner loop, i.e., when the region growing process reaches the adjacent part, there is a significant difference between d_{l+1}^i and d_l^i ; thus, when $|d_{l+1}^i - d_l^i|/d_{l+1}^i > \Phi$, the region growing process should be terminated. The obtained S_{layer_l} can be considered as a subset of the region segmentation.

However, the computation of an appropriate value of Φ is a non-trivial task due to the various shapes and scales of the parts. Therefore, the linear least-squares method [51] is utilized to predict the value of d_{l+1}^p of the next layer according to the values of d^i of the previous layers, and the computation of d_{l+1}^p takes into account both the shape and scale information of each part. Specifically, we define $d^i = X\hat{w}$, where $X^T = \begin{pmatrix} 1 & 2 & \dots & l \\ 1 & 1 & \dots & 1 \end{pmatrix}$, $\hat{w} = (w; b)$. Then the least-squares method is utilized to estimate $\hat{w}$,

$$\begin{aligned} \hat{w}^* &= \underset{\hat{w}}{\operatorname{argmin}} E_{\hat{w}^*} \\ &= \underset{\hat{w}}{\operatorname{argmin}} (d^i - X\hat{w})^T (d^i - X\hat{w}). \end{aligned} \quad (3)$$

Algorithm 2 skeletal curves extraction**Input:** P, p_c , endpoints set p^P , parameter Φ **Output:** skeletal curve set C

```

1: function EXTRACTION( $P, p_c, p^P, \Phi$ )
2:    $V \leftarrow \emptyset, C \leftarrow \emptyset, i \leftarrow 0$ 
3:   while  $size(V) < size(P)$  do
4:     if  $i < size(p^P)$  then  $p_x \leftarrow p_i^P$ 
5:     else  $p_x \leftarrow \text{GetStartPoint}(P, V, p_c)$ 
6:     end if
7:      $i \leftarrow i + 1, l \leftarrow 0, c = \emptyset$ 
8:     while  $p_x$  do
9:       if  $l = 0$  then  $S_{seg_1}^i, q_1^i, d_1^i \leftarrow p_x$ 
10:      end if
11:       $S_{seg_{l+1}}^i, q_{l+1}^i, d_{l+1}^i \leftarrow S_{seg_l}^i, q_l^i, d_l^i$ 
12:      if  $l > 2$  then
13:         $d_{l+1}^p \leftarrow \text{predict}(\{d_1^i, d_2^i, \dots, d_l^i\})$ 
14:         $\lambda \leftarrow |d_{l+1}^i - d_l^i|/d_{l+1}^i$ 
15:      else  $\lambda \leftarrow 0$ 
16:      end if
17:      if  $\lambda < \Phi$  then  $c.add(q_l^i), V.add(S_{seg_l}^i)$ 
18:      else break
19:      end if
20:       $l \leftarrow l + 1$ 
21:    end while
22:     $C.add(c)$ 
23:  end while
24: end function
25: function GETSTARTPOINT( $P, V, p_c$ )
26:    $U \leftarrow \{p_u | p_u \in P, p_u \notin V\}$ 
27:    $p_u^x \leftarrow \max(\|p_u - p_c\|)$ 
28:   if  $Knn_{p_u^x}(\delta\ell) > 10$  then return  $p_u^x$ 
29:   else  $\text{GetStartPoint}(P, V.add(p_u^x), p_c)$ 
30:   end if
31: end function

```

Subsequently, by setting $\frac{\partial E_{\hat{w}}}{\partial \hat{w}} = 2X^T(X^T\hat{w} - d^i) = 0$; and $\hat{w}$ is obtained. Thus, d_{l+1}^p is estimated by $d_{l+1}^p = w(l+1) + b$. Therefore, the value of $|d_{l+1}^i - d_l^i|/d_{l+1}^i$ will not largely change with different shapes and scales of different parts, and loop break threshold Φ is more stable. Therefore, when d_{l+1}^p and d_{l+1}^i meet Eq. (4), the growing of the current region is terminated.

$$|d_{l+1}^i - d_{l+1}^p|/d_{l+1}^i > \Phi. \quad (4)$$

Then, a new initial seed (the endpoint) is selected to divide the layers to process another part. Ideally, each part has an endpoint; however, some parts that are surrounded by the other parts will have no endpoints, and will not be processed by the procedure described previously. For example, consider the octopus models in Fig. 13. Hence, among the points that have not been labeled as “visited”, the unvisited point p_x that is farthest from the center point p_c of the OBB of the P is taken as the initial seed to continue the region growing procedure. When there is no new initial seed and all points on the point cloud have been visited, the algorithm is terminated. The region growing pseudocode is shown in Algorithm 2.

Small Radius Constraint. When dealing with a model like the hand shown in Fig. 6(k), a too-large value of radius Δ will affect the extraction effect. The gap between different fingers will be smaller than radius Δ , which will lead to errors in region growing. To prevent this, Δ is refined. The l cycle in the inner loop is divided into τ times, the small radius of each time is $\rho = \Delta/\tau$, and the points are searched for τ times to obtain next segment $S_{layer_{l+1}}^i$. In the process to obtain the next segment $S_{layer_{l+1}}^i$, the distance between the outliers and the points on the model is larger than the small radius ρ , while

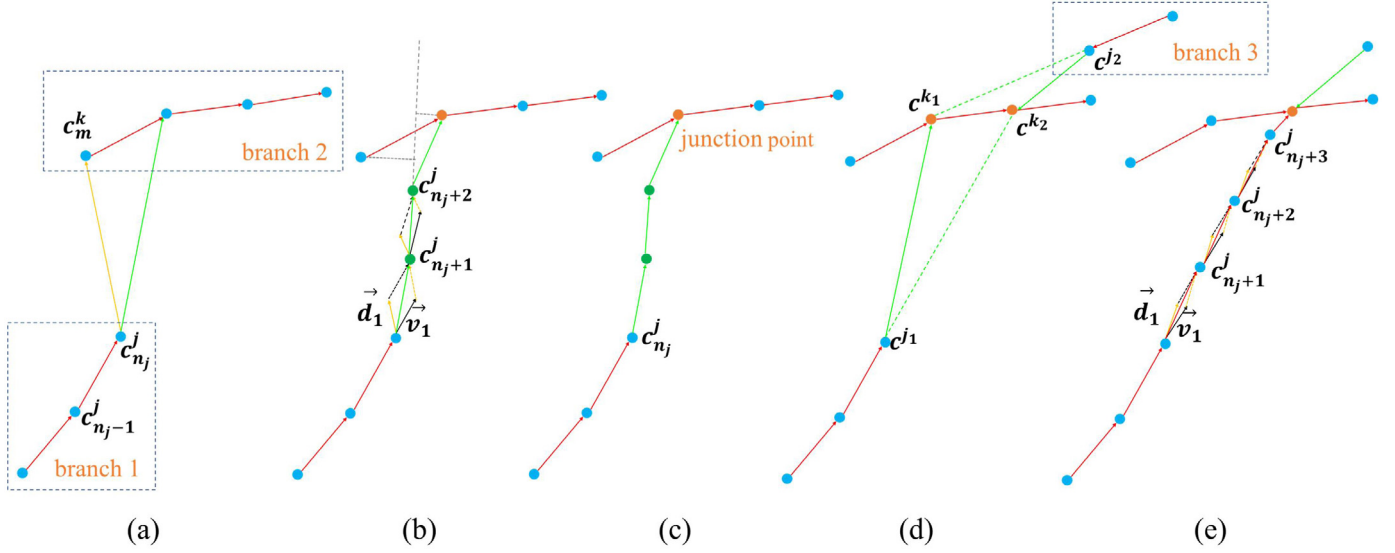


Fig. 5. The process of locating junction points. (a), (b) and (c) show the moving points driven by the multi-vectors, (d) shows the process of topology optimization, and (e) is the final result.

the distance between the points on the model is relatively smaller than ρ , therefore the outliers are excluded from the next segment $S_{layer_{i+1}}^i$, which can effectively reduce the influence of the outliers on skeletal point generation. The outliers are marked in gray in Fig. 2(n), which are not considered when computing the skeletal points. Since τ represents the times of the loop, a larger τ means more inner loops, which affects the efficiency. However, a smaller τ may induce region growing error. We set for all tests in this work, $\tau = 4$.

3.4. Construct the curve skeleton

The skeletal points were generated in a linear sequence in the previous stage, as they were generated during the region growing process. Therefore, the skeletal curves of each part can be formed by directly connecting the skeletal points in the linear sequence; thus, the skeletal curves are formed as the branches of the final curve skeleton, as shown in Fig. 2(h).

Given the branches, the final skeleton is constructed in this stage, which completes two main tasks. First, the proper junction points for connecting branches are located by the proposed MVMD method. Second, the global topology of the skeleton is optimized by reducing the redundant junction points and then re-connecting the branches using the proposed MVMD method. Furthermore, the distribution of skeletal points is non-uniformly sampled due to the shape differences between different parts of the models, which considerably affects the smoothness of the skeletal curves. We therefore apply an interpolation method based on the 1-4 splitting operator [49] to smooth the skeletal curves and subsequently resample to make sure the skeletal points are uniform. Finally, a complete and smooth curve skeleton can be obtained.

3.4.1. MVMD for locating junction points

The junction points are referred to as the points that connect at least two branches of the skeleton, and are crucial for the entire shape of the final skeleton. Therefore, both the distance and direction vectors are considered as multi-vectors for the selection of the junction points, i.e., multi-vectors are formed according to the nearest distance and direction of the current branches, both of which determine the location of the junction points. The details are as follows.

For each branch, the last skeletal point $c_{n_j}^j$ that is the farthest from the endpoint p_j^p is selected as the target point, as shown in Fig. 5(a). $c_{n_j}^j$ represents the last skeletal point on the j th branch, and n_j denotes

the number of skeletal points on the j th branch. The multi-vectors are then computed. The first vector is $\vec{v}_0$, which is formed by the points $c_{n_j}^j$ and $c_{n_j-1}^j$, among which $c_{n_j-1}^j$ is the adjacent skeletal point of $c_{n_j}^j$ on the j th branch; thus, the direction of the j th branch can be retained. The other vector $\vec{d}_0$ is formed by points $c_{n_j}^j$ and c_m^k , among which c_m^k is the m th skeletal point on the k th branch (there are n_k points on the k th branch) that meets $c_m^k = \min_{x=1 \sim n_k} \|c_{n_j}^j - c_x^k\|$. The definitions of $\vec{v}_0$ and $\vec{d}_0$ are as follows.

$$\begin{aligned}\vec{v}_0 &= c_{n_j}^j - c_{n_j-1}^j \\ \vec{d}_0 &= c_m^k - c_{n_j}^j.\end{aligned}\quad (5)$$

Given $c_{n_j}^j$, $\vec{v}_0$ and $\vec{d}_0$, the next point $c_{n_j+1}^j$ is computed using Eq. (6), which is a linear combination of $\vec{v}_0$ and $\vec{d}_0$, as shown in Fig. 5(a) and (b).

$$\begin{aligned}\vec{v}_t &= c_{n_j+t}^j - c_{n_j+t-1}^j \\ \vec{d}_t &= c_{m_t}^{k_t} - c_{n_j+t}^j \\ c_{n_j+t+1}^j &= c_{n_j+t}^j + \mu (\alpha \cdot \vec{v}_t + (1 - \alpha) \cdot \vec{d}_t),\end{aligned}\quad (6)$$

where α is balancing factor among $\vec{v}_t$ and $\vec{d}_t$ by using the concept of momentum.

By repeating the preceding procedure, the points are moved forward as driven by the multi-vectors, as shown in Fig. 5(b). The repeating process terminates when $c_{n_j+t+1}^j$ meets $\|c_{n_j+t+1}^j - c_{m_t}^{k_t}\| < \frac{\epsilon}{20}$, which indicates two branches have been connected (as shown in Fig. 5(b) and (c)), or when the moving step reaches 10, which means the two branches cannot be connected because they are too far away.

The main parameters controlling the moving process are μ and α . The $\mu = \frac{\epsilon}{40}$ is set by default, and the value of α is computed by Eq. (7). Moreover, t is the moving step number, and the value of α decreases with the increase of t .

$$\alpha = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(t/5)^2}{2}\right), \quad (7)$$

thus, as t gradually increases, α decreases; correspondingly, the weight of $\vec{v}_t$ decreases and the weight of $\vec{d}_t$ increases. In other words, in the previous steps, the direction information largely affects the location of $c_{n_j+t+1}^j$, whereas in the last several steps, the location of $c_{n_j+t+1}^j$ is mainly affected by the distance information. The step number t is used as a parameter for Standard Normal Distribution and α can be calculated

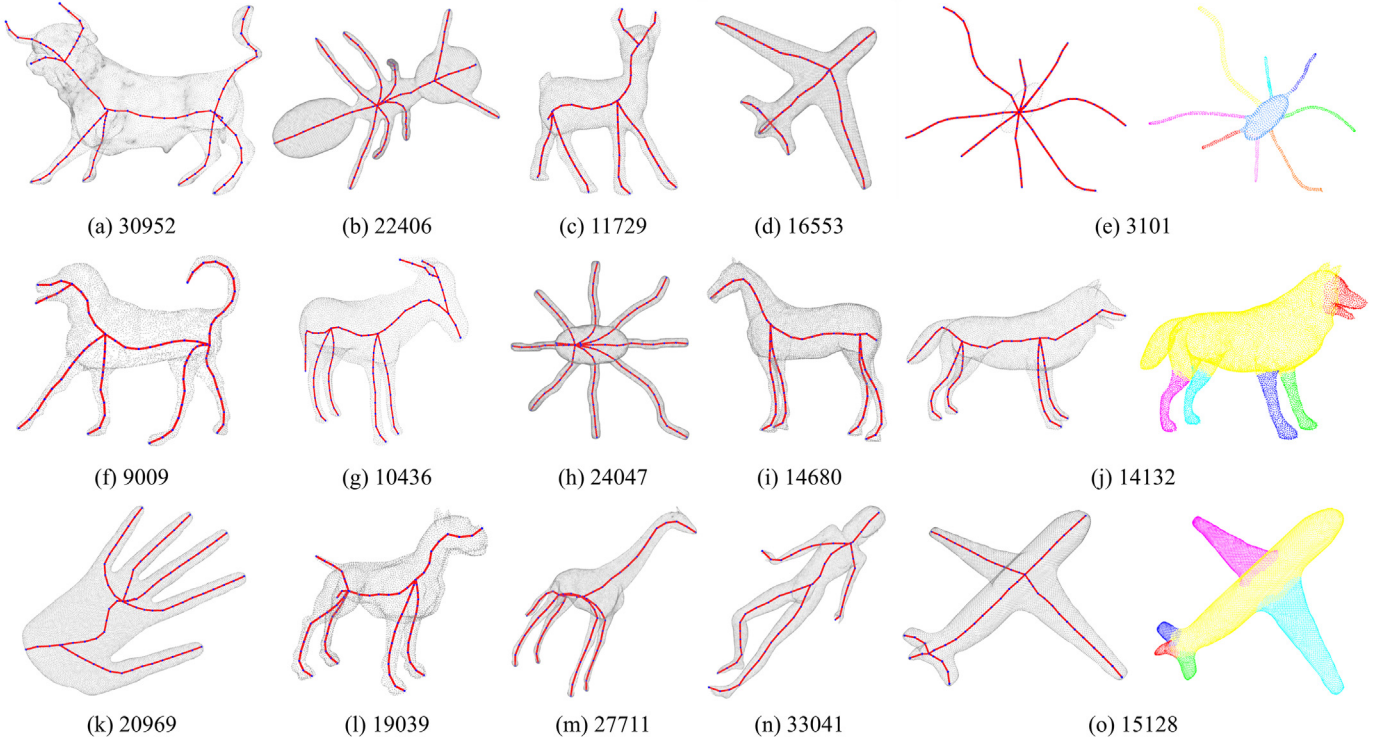


Fig. 6. The curve skeleton extracted by EPCS from different point cloud models. The number below each model is the number of points in the model. The last column is the model segmentation result corresponding to the model in the previous column.

by Eq. (7). Via the momentum-driven dynamic coefficient setting, the MVMD method can locate more appropriate junction points.

By performing the MVMD method, the independent branches are connected, and a complete skeleton can be obtained. Furthermore, the resulting skeleton is smoother because the branches are connected by interpolating points, as shown in Fig. 5(a), instead of by directly generating a straight line to connect the branches.

3.4.2. Global topology optimization

The previous stages can yield a complete skeleton for the given point cloud; however, a problem remains because several redundant junction points may also be generated. This is because the junction points are located by only considering the local direction and distance information. The global topology of the skeleton therefore requires optimization.

To optimize the global topology of the skeleton, all the junction points are first checked by computing the Euclidean distances between their adjacent junction points, e.g., junction points c^{k_1} and c^{k_2} in Fig. 5(d). For the sake of a clear representation, the adjacent junction point pair $\langle c_{n_j}^j, c_m^k \rangle$ is denoted as $\langle c^j, c^k \rangle$. For c^{k_1} and c^{k_2} , there are two pairs, namely $\langle c^{j_1}, c^{k_1} \rangle$ and $\langle c^{j_2}, c^{k_2} \rangle$. If the Euclidean distance between them is less than the topology optimization parameter ζ , one of the two junction points is redundant, and one is removed using the following procedures; otherwise, these two junction points should be retained.

For a junction point pair that has redundant junction points, Eq. (8) is used to identify the redundant junction points, and the branch is then connected with the remaining junction point, as shown in Fig. 5(d).

$$\begin{cases} d_1 = \|c^{k_1} - c^{j_1}\| + \|c^{k_1} - c^{j_2}\| \\ d_2 = \|c^{k_2} - c^{j_1}\| + \|c^{k_2} - c^{j_2}\|, \end{cases} \quad (8)$$

if $d_1 < d_2$, c^{k_1} is considered a common junction point among c^{j_1} and c^{j_2} , and vice versa.

The topology optimization process removes the redundant junction points, as shown in Fig. 5(e); however, it also results in some

independent branches that were previously connected with the redundant junction, e.g., branches 1 and 3 in Fig. 5(e). Therefore, these independent branches are re-connected with the remaining nearest junction points using the MVMD method described in Section 3.4.1. By using the MVMD method, which employs the interpolation operation to complete the connection task, a complete and smooth skeleton can be constructed.

Specifically, the goal described in Section 3.4.1 is to locate the junction point; here, the junction point is already known and denoted as c_b^a . Therefore, instead of c_k^m and $c_{m_i}^k$, only c_b^a needs to be used to rewrite $\vec{d}_0 = c_b^a - c_{n_j}^j$ and $\vec{d}_1 = c_b^a - c_{n_{j+t}}^j$. With the help of $c_{n_{j+t+1}}^j = c_{n_{j+t}}^j + \mu(\alpha \cdot \vec{v}_t + (1 - \alpha) \cdot \vec{d}_t)$ to complete the interpolation process, the obtained $c_{n_{j+t+1}}^j$ is considered as the skeletal point, as shown in Fig. 5(e).

3.4.3. Skeleton uniform and smooth

Due to the shape differences between different parts of the model, the distribution of the skeletal points is non-uniform, resulting in not smooth skeletal curves. By interpolating and resampling the skeletal curves, the distribution of the skeletal points can be uniform, which also makes the skeletal curves smoother. Thus, an interpolation method based on the 1-4 splitting operator [49] is adopted, and we set the distance regularization parameter $\eta = 0.1$ to keep skeletal points not to be too close.

Specifically, for a skeletal curve, we interpolate until the distance between any two skeletal points of the branch is less than $\frac{1}{5}\eta$, as shown in Fig. 2(l). Then we down sample the skeletal points using η as the sampling density, finally, all skeletal points are reconnected and a complete EPCS can be obtained, as shown in Fig. 2(m).

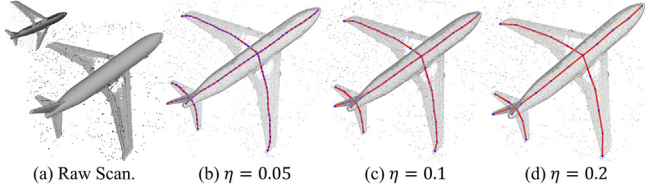
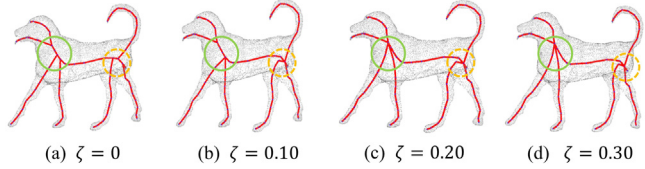
4. Result and analysis

To demonstrate the effectiveness and efficiency of the proposed EPCS algorithm, experiments were performed on various types of point clouds, including models with various amounts of noise (Figs. 2, 7, 10,

Table 1

Default settings used for user controllable parameters.

Parameter	Φ	η	$\Delta = \tau \times \rho$	ζ	ϵ	δ	ϕ
Default Value	0.2	0.1	$0.1 = 4 \times 0.025$	0.2	0.02	0.0625	5

**Fig. 7.** η determines the spacing between the skeletal points on the curve skeleton.**Fig. 8.** ζ determines the degree to which the junction point is removed.

11, and 12), models with missing data (Figs. 1, 2, 7, 9, 10, 11, and 14), models with outliers (Figs. 2, 7, 10, 11, and 13), raw scan point clouds (Figs. 2, 7, 10, and 11), and the real scanned Terracotta dataset (Figs. 1, and 14). Furthermore, to highlight the advantages of the proposed method, it was also compared with several state-of-the-art methods, including the LBC [9], L_1 -medial [6], MdCS [10] and GC [11] methods. The codes of the LBC [9], L_1 -medial [6] and MdCS [10] methods and the executable program of GC were provided by the authors, and all the parameters were fine-tuned. To perform the experiments, a computer equipped with an AMD 3700X CPU, an Nvidia RTX 2060S GPU, and 32 GB of RAM was used. LBC [9] and MdCS [10] were implemented in MATLAB, L_1 -medial [6] and EPCS in C/C++, and GC in Java. All of them ran on CPU. Furthermore, the Robustness Analysis did not involve the GC [11], because its executable program lacked an interface for exporting the skeleton points.

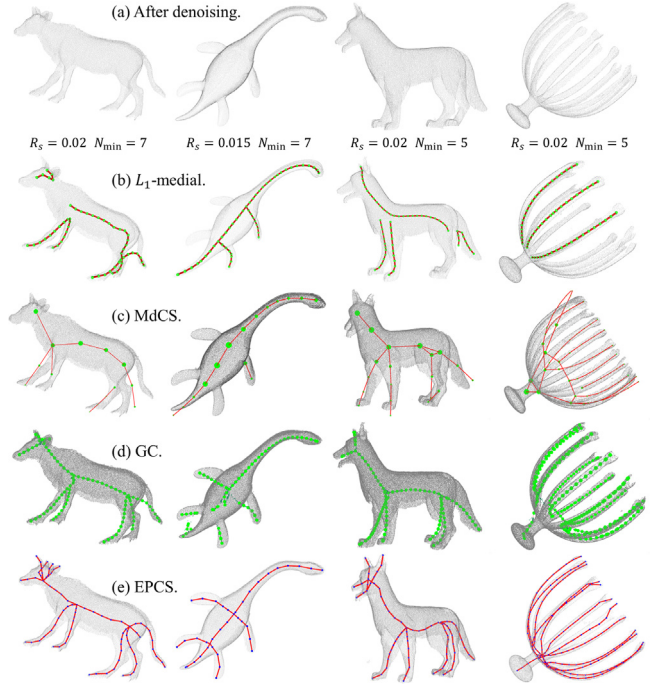
4.1. Parameters

The default parameters of the proposed algorithm are listed in Table 1. Distance regularization parameter η determines the spacing between the skeletal points on the curve skeleton, the larger η is, the farther between the skeletal points, and the sparser the skeletal points. As shown in Fig. 7.

Loop break threshold Φ is an important parameter for controlling the region growing process. With the help of the least-squares method, Φ converges to 0.2 for different model. Moreover, topology optimization parameter ζ , which is used for merging junction points. When ζ increases, the nr. of junction points will decrease, and vice versa. From Fig. 8 (a–c), it can be seen that when ζ increases from 0 to 0.1, the three junction points marked in the yellow dotted circle are merged together, and when increasing from 0.1 to 0.2, the two junction points marked in the green circle are also merged together. The distance between the junction points is less than 0.2, so ζ increases from 0.2 to 0.3, no new merging occurs. The parameters ϵ , δ , and ϕ are fixed in our algorithm.

4.2. Raw scans results

This section compares the actual performance of several algorithms on the raw scans. Extracting curve skeletons from raw scan models is challenging because raw scan models exhibit non-uniform sampling,

**Fig. 9.** The first row shows the point clouds after denoising, and the following four rows show the curve skeleton results of LBC [9], L_1 -medial [6], MdCS [10], GC [11] and EPCS, respectively.

contain holes, noise, and outliers. In Fig. 10(b), the dark area indicates missing data caused by holes, and the outliers and noise are surrounding the model.

It can be seen from Fig. 10 that the LBC [9] algorithm is sensitive to outliers; although it can deal with holes, under the influence of both noises and outliers, the curve skeleton of LBC [9] is no longer available. The L_1 -medial skeleton [6] cannot converge due to the growing radius of the neighborhood ball, resulting in the absence of skeletal branches. Outliers affect the quality of MdCS [10] since they considerably affect the symmetry of the topology check, manifesting as missing branches. Although the GC [11] uses WLOP [15] to reduce the influence of outliers, it still produces topological errors on the limbs.

For EPCS, the impact of missing data on the RCS method is limited because the shift is also notably large at the endpoints, as illustrated in Fig. 3. Since EPCS predicts the curve skeleton topology in advance via RCS method for endpoints detection and avoids the lack of skeleton branches, the extracted curve skeletons are more complete than other methods. In skeletal points extraction, by setting a small radius constraint on the region growing process, outliers can be excluded and the region growing process starting from the outliers cannot be initiated, thus avoiding the influence of outliers on the extraction of skeletal points. Furthermore, since the principal direction of the point set is calculated in the construction of OBB, the interference of noises and outliers can also be reduced to some extent, thereby reducing the disturbance to skeletal points. Additionally, though there are holes that cause data missing in raw scan point clouds, since OBB is not sensitive to missing data, even if there are holes that cause data missing in raw scan point clouds, it will not have a large impact on OBB, thus ensuring better centering of skeletal points. In order to further reduce the influence of noises, outliers and missing data on the curve skeleton, the MVMD method is designed to globally and locally optimize the extracted skeleton, and a smoothing method is also utilized to smooth the skeleton curves, thereby improving the quality of the curve skeleton.

Afterward, we employ the denoising method on the raw scans, which are shown in the first row of Fig. 9. The denoising method

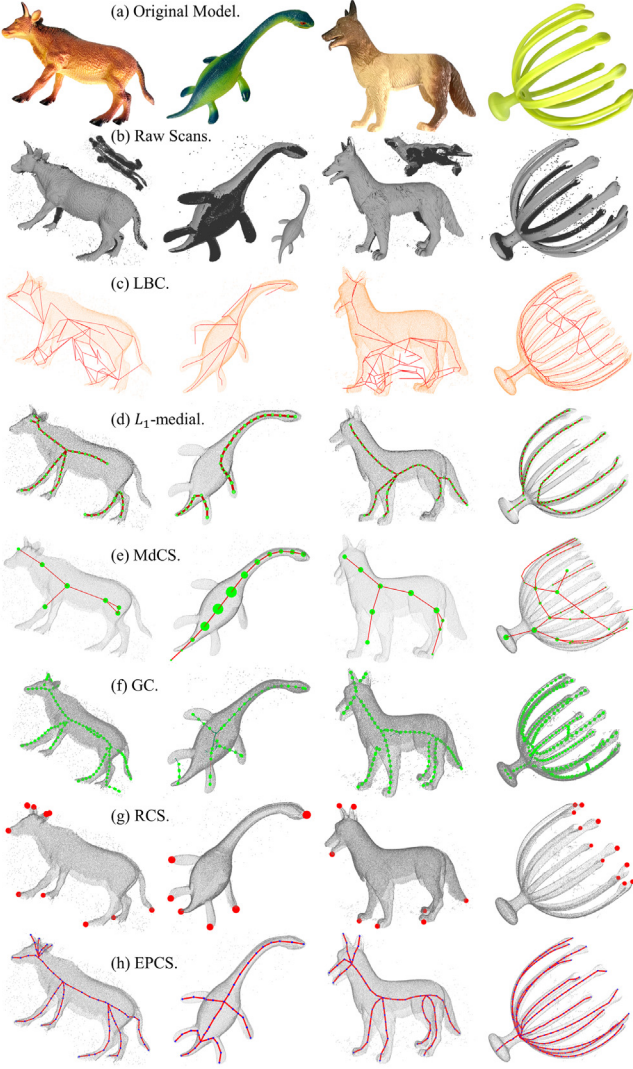


Fig. 10. The skeleton of raw scans with missing data, noises, and outliers. The first row is the original models, the second row is the raw scans, the dark area indicates missing data, and the points surrounding the model are outliers and noise. The next four rows are the results of LBC [9], L_1 -medial [6], MdCS [10], GC [11], the seventh row shows the endpoints detected by RCS method and the eighth row shows the curve skeleton extracted by EPCS, respectively.

removes outliers and noise from point clouds by checking whether the number of points within a neighborhood search radius R_s of any point exceeds a threshold N_{min} . The values of neighborhood radius R_s and threshold N_{min} are also listed in Fig. 9. Moreover, L_1 -medial [6] improved the curve skeleton extraction by reducing branch loss, whereas MdCS [10] extracted more detailed branches. Furthermore, GC [11] corrected the branch topology errors on the limbs, and EPCS extracted more reasonable topology structures on the denoising models.

4.3. Robustness analysis

This section presents results on models disturbed by missing data, noise, and outliers, to verify the robustness of the proposal method.

4.3.1. Quantitative analysis

Missing data. Fig. 11 presents the results of the four algorithms when dealing with point clouds with missing data. In the experiment, f is the number of raw scan frames, and $f = 2, 3, 4, 5$. When f increases, the horse model is gradually complete.

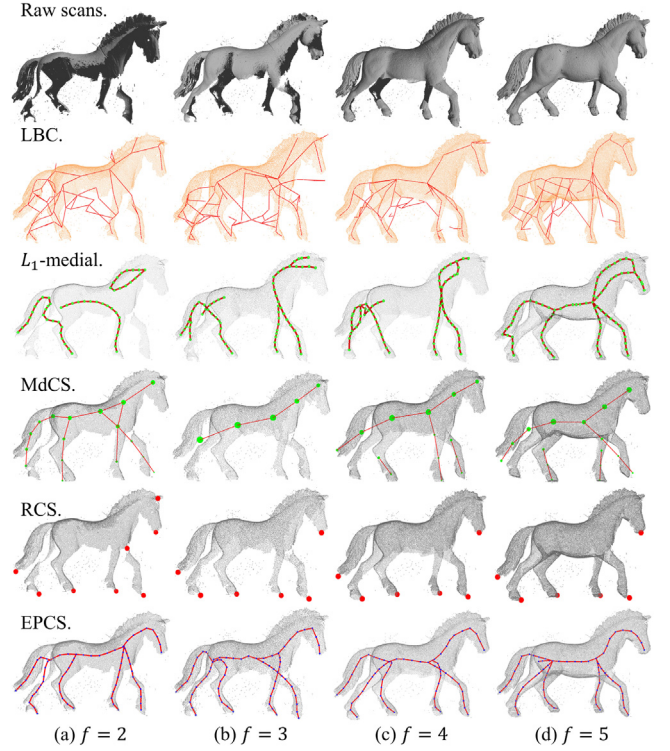


Fig. 11. The skeleton of raw scans with missing data. f increases, the model is gradually complete. The first row is the models, the next five rows are the results of LBC [9], L_1 -medial [6], MdCS [10], the endpoints detected by RCS method and the curve skeleton extracted by EPCS, respectively.

As shown in Fig. 11, the L_1 -medial [6] and MdCS [10] methods yielded inadequate solutions for the multi-branch model, which indicates that the LBC [9], L_1 -medial [6], MdCS [10] methods all encountered challenges in skeleton extraction on multi-branch models with missing data. LBC [9] cannot handle models with outliers, thus unavailable skeletons are extracted. With f increasing, the L_1 -medial skeleton becomes more and more complete, and when $f = 5$, although the skeleton is extracted, a connection error between the tail and legs is produced. MdCS [10] performs poorly due to noises and outliers interfering with topology checking.

Since the influence of noises and outliers are taken into account in the RCS method, when $f > 2$, the detected endpoint is ideal and can be used as a seed point for skeletal points generation. Although the invalid endpoints are detected at the horse's ear, the final skeleton extraction result is not affected. This is because the adaptive radii r_i^p of the invalid endpoints are significantly larger than the valid endpoints, making the region growing process starting from the invalid endpoints initiate later; however, the neighbors of the invalid endpoints have been visited by the other region growing process, and thereby the region growing starting from the invalid endpoints will not initiate, and no skeletal points are generated.

Noise. For the experiment on models with Gaussian noises, the standard deviation of Gaussian noise is defined as σd_{min} , where d_{min} is the minimum distance between points in the point cloud model. The Gaussian noise disturbance experiment is expressed as $Noise(\sigma)$, and $Noise(\sigma = 0)$ refers to no Gaussian noise added to the model. The experimental results are shown in Fig. 12.

It can be seen from Fig. 12 that when $Noise(\sigma = 0)$, the LBC [9], L_1 -medial [6], MdCS [10], and EPCS methods can extract a complete and correct curve skeleton; however, when $Noise(\sigma = 2)$, the LBC [9] method produced a loop structure on the chest of the human model, the L_1 -medial [6] method produced an incomplete curve skeleton because the body and the head were not connected, and the MdCS [10]

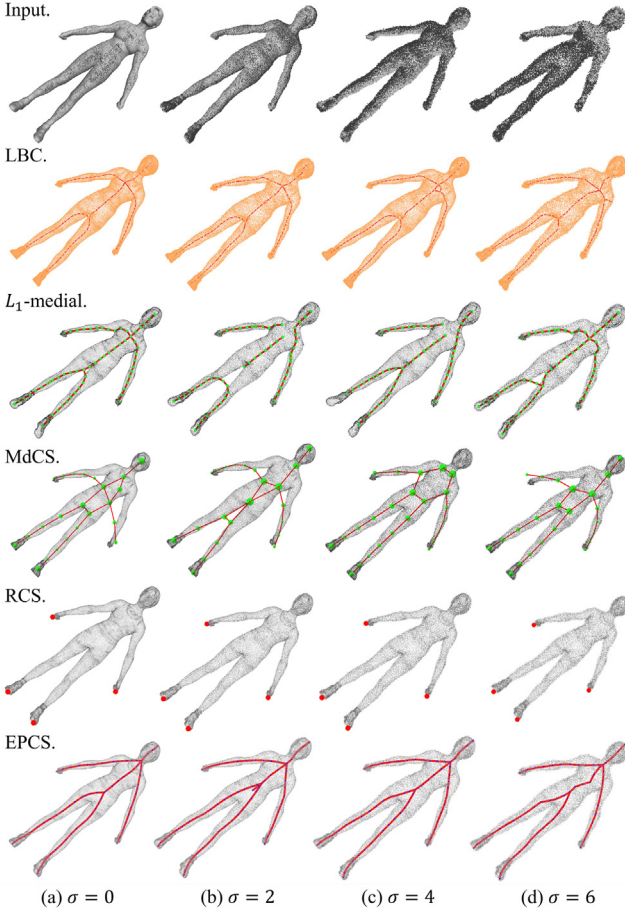


Fig. 12. Curve skeleton of models disturbed by Gaussian noises, the larger the σ , the more serious the noise. From the second row to the bottom are the skeleton extraction results of LBC [9], L_1 -medial [6], MdCS [10], the endpoints detected by RCS method, and the curve skeleton extracted by EPCS, respectively.

method produced the worst curve skeleton with incorrect branches and connection relationships. When $Noise(\sigma = 4)$ occurred, the connectivity among the skeleton curves of the L_1 -medial [6] method continued to deteriorate, and the MdCS [10] method yielded two adjacent skeleton curves on the abdomen of the human model. With the further increase in the scale of noise, such as $Noise(\sigma = 6)$, the LBC [9] method resulted in redundant branches. The L_1 -medial [6] method was characterized by distortion in the processing of the legs. The MdCS [10] method produced the same error as in the case of $Noise(\sigma = 2)$; the junction points of the skeleton curve were selected incorrectly. However, the proposed EPCS algorithm achieved a satisfactory result despite the interference of these types of noise.

Outliers. In the experiments on models with outliers, o was utilized to represent the number of outliers in the model. o is the ratio of the number of outliers added to the models; specifically, $o = 2, 4, 6$, means $2\% \times I$, $4\% \times I$, $6\% \times I$ outliers are added to the point cloud, where I represents the number of the points on the original model. The outliers were restricted to a space 1.2 times larger than the point cloud model space. As shown in Fig. 13, the LBC [9] method cannot handle models with outliers and almost no effective results are obtained, this is because when the distance between the outliers and the original point of the model is too close, the Delaunay triangulation algorithm used by the LBC [9] method may fail. Therefore, for this experiment, only the L_1 -medial [6], MdCS [10] and the proposed method were evaluated.

It can be seen from Fig. 13, even when $o = 0$, the skeleton extracted by the L_1 -medial [6] method exhibits topological errors, although the skeleton was successfully extracted on the feet of the octopus, which

reflected better performance than for the horse model shown in Fig. 11. This is because the change in the radius of the local neighborhood ball between the horse's leg and body was greater than the radius of the local neighborhood ball between the leg of the octopus and the body, which eased the contradiction between the initial local neighborhood value and the growth rate to a certain extent. Regarding the MdCS [10] method, the outliers also caused more serious interference to the topology check. In particular, for the topology check, principal component analysis (PCA) is primarily used to detect the symmetry of the transport plan between points, and outliers will greatly interfere with the results of the PCA algorithm; thus, the topology error of the MdCS [10] method gradually became serious. For the L_1 -medial [6] method, with the increase of o , the outliers introduced interference to the increase of the local neighborhood ball radius, resulting in the gradual distortion of the extracted skeleton. Furthermore, due to the existence of the sampling process, the possibility of using outliers as sampling points progressively increased, and because there were no other points around the outliers, they could not be transmitted to other branches by the algorithm, resulting in a greater lack of skeletal branches.

For the EPCS method, outliers leading to introduce challenges to the endpoint recognition algorithm, resulting in more endpoints that affect subsequent steps. However, since the radius of the endpoint is adaptively computed, the radius of the outliers is infinitely close to 0. Furthermore, even if an outlier is used as a seed point, the region growing process cannot be started due to the absence of neighboring points. Therefore, even if an outlier poses a challenge to endpoint recognition, it will not affect the extraction of the curve skeleton. However, when the outliers are located close enough to the point cloud, slight disturbance to the curve skeleton will occur, but these changes will not considerably affect the quality of the skeleton. In summary, the proposed algorithm is the least affected by outliers, and a valid curve skeleton can still be extracted when $o = 6$.

Quantitative Analysis. To quantitatively analyze the quality of curve skeleton extraction results from point clouds with different types of disturbance, mean curvature skeleton (MCS) [34] extracted from the clean mesh model is considered as the ground truth [10]. The Hausdorff distance is a commonly used metric for evaluating the differences between two skeletal curves [27], [10], we thus adopt two-sided Hausdorff distances to evaluate these methods.

Table 2 shows the detailed data of $h(A, B)$ and $h(B, A)$. $h(A, B)$ shows that the Hausdorff distance data of EPCS is significantly smaller than that of L_1 -medial [6] and MdCS [10], indicating that the skeleton extracted by EPCS is more complete and reasonable than that of L_1 -medial [6] and MdCS [10]. Moreover, Fig. 12 shows that the LBC [9] performs better when $\sigma = 0$ and $\sigma = 1$ than in L_1 -medial [6] and MdCS [10], which is consistent with that in Table 2.

The results of EPCS are all less than 0.1 in the three cases of $f = 6$, $\sigma = 0$ and $o = 0$, which indicates that EPCS is closer to the ground truth than the other three methods. When σ and o increase, $h(A, B)$ of EPCS are always in a relatively stable interval, which verifies the robustness to noise and outliers.

$h(B, A)$ shows the gap between the curve skeleton and ground truth extracted by several methods, ignores the absence of skeletal branches, and reveals the neutral difference of the skeletal points of several curve skeletons. It can be seen from Table 2 that the skeletal point of LBC [9] is closer to the ground truth under the premise that the skeleton can be obtained. EPCS is slightly worse than LBC [9] since the topology structure of EPCS near the junction points is generated by MVMD method, whereas both LBC [9] and MCS [34] are obtained by contraction, but EPCS still better than L_1 -medial [6] and much better than MdCS [10], furthermore, these conclusions are also proven in Table 2. When dealing with the raw scans (with noises, outliers, and missing data), with the increase of frames, the EPCS is closer to the ground truth.

Table 2
Quantitative comparison of curve skeleton results.

Horse model with missing data.									
A	B	$h(A, B)$				$h(B, A)$			
		$f = 2$	$f = 3$	$f = 4$	$f = 5$	$f = 2$	$f = 3$	$f = 4$	$f = 5$
MCS	L_1 -medial	0.344	0.302	0.243	0.281	0.189	0.416	0.253	0.331
	MdCS	0.309	0.385	0.360	0.347	0.228	0.177	0.181	0.166
	EPCS	0.172	0.207	0.266	0.056	0.164	0.243	0.201	0.125
Human model with noise.									
A	B	$h(A, B)$				$h(B, A)$			
		$\sigma = 0$	$\sigma = 2$	$\sigma = 4$	$\sigma = 6$	$\sigma = 0$	$\sigma = 2$	$\sigma = 4$	$\sigma = 6$
MCS	LBC	0.145	0.148	0.154	0.158	0.132	0.133	0.140	0.149
	L_1 -medial	0.155	0.151	0.148	0.143	0.190	0.186	0.182	0.171
	MdCS	0.217	0.201	0.282	0.202	0.280	0.287	0.313	0.364
	EPCS	0.093	0.072	0.071	0.071	0.184	0.186	0.189	0.191
Octopus model with outliers.									
A	B	$h(A, B)$				$h(B, A)$			
		$o = 0$	$o = 2$	$o = 4$	$o = 6$	$o = 0$	$o = 2$	$o = 4$	$o = 6$
MCS	L_1 -medial	0.529	0.560	0.486	0.712	0.155	0.144	0.219	0.297
	MdCS	0.440	0.442	0.578	0.440	0.509	0.487	0.489	0.489
	EPCS	0.091	0.108	0.087	0.088	0.261	0.264	0.243	0.187

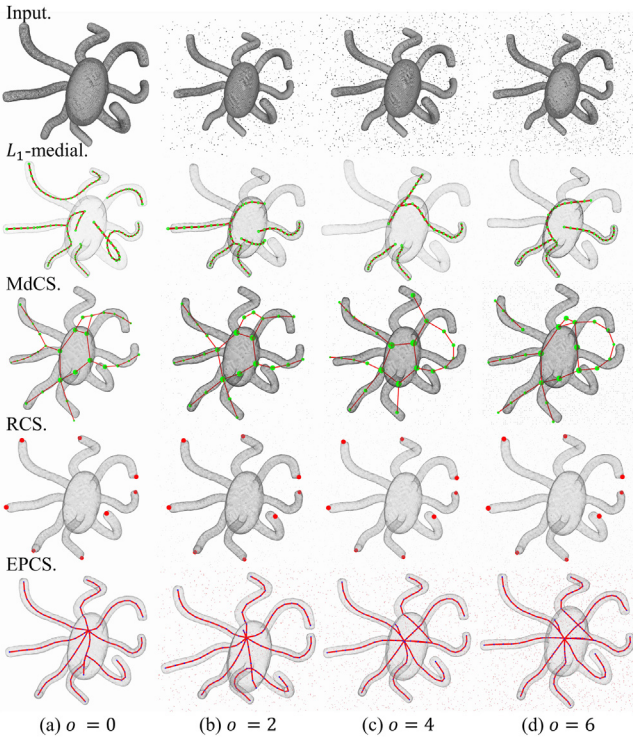


Fig. 13. The skeleton of the three methods when processing models with outliers. o is the ratio of the number of outliers added to the models, the larger o means more outliers. From second row to bottom are the results of L_1 -medial [6], MdCS [10], the endpoints detected by RCS method, and EPCS, respectively.

4.3.2. Efficiency

The efficiency of the proposed method was compared with that of the LBC [9], L_1 -medial [6], MdCS [10], and GC [11] methods in this section. The models shown in Fig. 6 were employed for testing. The scale of these models varies from 3101 to 33041. Table 3 presents the execution time consumed by the three baseline algorithms and the proposed algorithm.

It can be seen that the LBC [9] and GC [11] method achieved relatively stable efficiency, i.e., as the number of points increased, the time consumption steadily increased. The MdCS [10] method is implemented using MATLAB; thus, the execution time was found to be

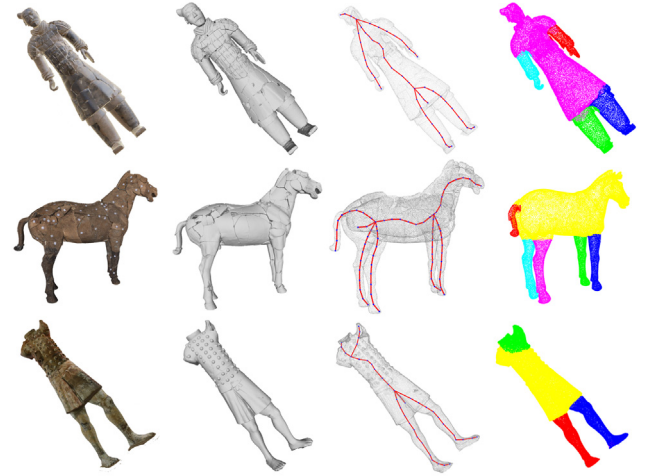


Fig. 14. The curve skeleton and model segmentation results of EPCS on the point clouds of broken Terracotta.

unsatisfactory. As shown in Fig. 6(f), when calculating the dog model, which contained only 9009 points, the MdCS [10] method still took 213 s. Although the L_1 -medial [6] method achieved better performance than LBC [9] and MdCS [10] methods, some extracted skeletal branches were missing, as shown in the second row in Fig. 13. As shown in Table 3, because EPCS contains no resampling process, all points of the model participate in the computation process; thus, an increase in the point number of the model directly leads to the increase of the time consumption.

4.4. Application

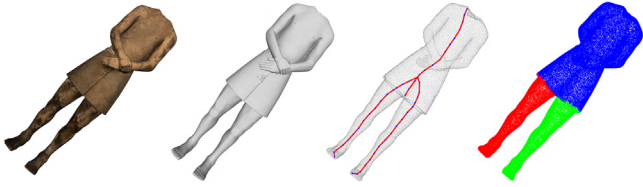
The curve skeletons of damaged cultural relics can visually display the entire pose of the relics; furthermore, they can also guide virtual and real restoration processes. However, the extraction of the curve skeletons of damaged cultural relics is a challenging problem. Even if broken cultural relics are manually restored, there may remain interference from large holes (missing data), noises, and outliers; this will lead to lower-quality models and ultimately result in the failure of many typical skeleton extraction algorithms.

The proposed EPCS method was applied on the models from Terracotta dataset, a real scanned dataset, and the method still achieved

Table 3

The computation time of LBC [9], L_1 -medial [6], MdCS [10], GC [11], and EPCS, unit (s). $Ratio_1$ is the time-consuming ratio between LBC [9] and EPCS, $Ratio_2$ is the ratio between L_1 -medial [6] and EPCS, $Ratio_3$ is the ratio between MdCS [10] and EPCS, $Ratio_4$ is the ratio between GC [11] and EPCS.

Models	#Vertices	Density	LBC	L_1 -media	MdCS	GC	EPCS	$Ratio_1$	$Ratio_2$	$Ratio_3$	$Ratio_4$
a-ox	30 952	15 079.3	132.630	33.030	5902.718	79.54	10.553	12.57	3.13	559.34	7.74
b-spider	22 406	9774.03	66.258	53.340	2084.692	45.36	5.793	11.44	9.21	359.86	7.83
c-deer	11 729	8423.7	33.124	18.670	302.403	32.68	2.615	12.67	7.14	115.64	12.50
d-plane	16 553	12 162	38.923	30.900	973.587	34.77	4.261	9.13	7.25	228.49	8.16
e-jellyfish	3101	1658.73	10.144	4.890	42.593	10.48	2.510	4.04	1.95	16.97	4.18
f-dog	9009	10 456.9	28.711	15.490	213.076	29.53	3.466	8.28	4.47	61.48	8.52
g-sheep	10 436	8788.03	29.472	17.580	271.891	31.87	2.989	9.86	5.88	90.96	10.66
h-spider	24 047	20 425	88.294	24.800	2370.769	44.20	7.046	12.53	3.52	336.47	6.27
i-horse	14 680	9347.26	34.460	18.120	497.692	28.94	3.268	10.54	5.54	152.29	8.85
j-wolf	14 132	6364.47	33.967	20.800	483.934	45.70	3.558	9.55	5.85	136.01	12.84
k-hand	20 969	17 003.8	91.344	76.250	1520.461	31.13	5.092	17.94	14.97	298.60	6.11
l-cheetah	18 159	14 711.5	46.356	16.420	1507.367	53.36	4.489	10.33	3.66	335.79	11.89
m-giraffe	27 711	34 106.6	104.881	15.800	2844.746	44.07	10.105	10.38	1.56	281.52	4.36
n-human	33 041	35 462.4	182.268	57.220	6258.986	54.11	13.081	13.93	4.37	478.48	4.14
o-plane	15 128	7461.05	36.318	21.530	527.902	41.76	4.082	8.90	5.27	129.32	10.23
mean	18 137	14 081.65	63.810	28.323	1720.188	40.50	5.527	11.54	5.12	311.22	7.33

**Fig. 15.** The cases EPCS cannot handle.

high performance. As can be seen from Fig. 14, when extracting the curve skeleton, the model segmentation results could also be obtained (as shown in the fourth column). It also can be seen from Fig. 14 that there were several holes in both the human figurine and horse models; however, the EPCS method still performed well. Regarding the Paoding warrior model in the last row, which is incomplete because both the arms are missing, the EPCS model still yielded a complete skeleton that was closer to the real pose of the model.

4.5. Limitations

The skeletonization of point clouds is usually an ill-posed problem [6], [10], especially in the case of missing data. When there is too much missing data or serious noises, the EPCS may fail. Moreover, the EPCS method cannot handle the case in which the branches of the model are stuck together, due to the characteristic of the region growing process. As shown in Fig. 15, the EPCS method failed to extract the curve skeleton of the arm.

5. Conclusion

In this study, the novel RCS method was designed for identifying the endpoints which are regarded as the seed points of region growing. Then the region growing was applied to complete the task of curve skeletal point computation. The complete topology of the curve skeleton was subsequently obtained using the MVMD method. Therefore, a curve skeleton extraction method, namely the EPCS, was proposed. The results of experiments demonstrate that the EPCS method is robust to unorganized, non-uniform sampling and un-oriented point cloud models with missing data, noises, and outliers. The EPCS can be applied to skeleton extraction on the models of broken cultural relics, highlighting the utilization of the proposed method. In light of the demand for real-world point cloud processing such as object recognition and classification, we further propose a novel deep network which uses EPCS as input to jointly complete these tasks.

CRedit authorship contribution statement

Chunhui Li: Writing – original draft, Conceptualization, Methodology, Software, Validation. **Mingquan Zhou:** Supervision, Revision suggestions. **Guohua Geng:** Supervision, Revision suggestions. **Yifei Xie:** Validation, Visualization. **Yuhe Zhang:** Funding acquisition, Supervision, Methodology, Writing – review & editing. **Yangyang Liu:** Funding acquisition, Writing – review & editing.

Declaration of competing interest

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Data availability

Data will be made available on request.

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